

Evaluating the Reliability of Gradient Boosting Models for Health Insurance Fraud Detection: A Quantitative Black-Box Explainability Analysis

Parsa Rassekhi¹, Dr. Seyed Hossein Iranmanesh²

¹ Department of Industrial Engineering, University of Tehran, Tehran, Iran

² Department of Industrial Engineering, University of Tehran, Tehran, Iran

Abstract

Health insurance fraud imposes substantial financial burdens on insurance systems worldwide, yet existing machine learning approaches report feature importance as a surrogate for explanation quality without rigorously quantifying whether those explanations are trustworthy. This study addresses this gap by evaluating explanation reliability across six complementary dimensions applied to three Bayesian-optimized gradient boosting architectures: LightGBM, XGBoost, and CatBoost, using a real-world Chinese health insurance claims dataset. Explanation evaluation encompasses fidelity analysis, stability assessment under Gaussian perturbations, counterfactual reasoning via Bayesian optimization, and error profiling. A novel Explanation Reliability Score (ERS) aggregates these dimensions into a single interpretable metric, revealing that CatBoost achieves the highest explanation trustworthiness (ERS=0.805), despite not achieving the highest predictive performance — a finding validated through Monte Carlo simulation across 1,000 weight perturbation scenarios. Moreover, PSI analysis indicates that static probability thresholds may lead to unstable alert volumes despite consistently high AUC. Ultimately, these findings demonstrate that selecting models based solely on predictive performance yields suboptimal deployment decisions, and incorporating quantitative explanation metrics provides the critical risk-management insights necessary to deploy trustworthy AI in high-stakes insurance auditing.

Keywords: : Health Insurance Fraud Detection; Gradient Boosting Decision Trees; Bayesian Optimization; SHAP; Explainable AI; Monte Carlo Simulation

1. Introduction

Healthcare insurance fraud represents one of the most economically damaging forms of financial crime in modern healthcare systems, diverting resources away from legitimate patient care and undermining the long-term sustainability of public and private insurance programs. Globally, fraudulent claims account for an estimated 3–10% of total healthcare expenditure, imposing annual financial burdens measured in hundreds of billions of dollars across major economies [1, 2]. In China specifically, the rapid expansion of the national medical insurance system has created both heightened fraud exposure and unprecedented data availability for computational detection, making the development of robust, scalable detection methodologies an urgent research priority [3].

The evolution of fraud detection has progressed through three broad technological generations. Rule-based expert systems, while interpretable, lack adaptability to evolving fraud tactics and suffer from high false positive rates as fraudsters learn to circumvent fixed thresholds [4]. Statistical anomaly detection methods improved sensitivity but struggled with the high-dimensional, heavily imbalanced claim datasets characteristic of national insurance systems [5]. The contemporary paradigm centers on supervised machine learning, with gradient boosting architectures — including LightGBM, XGBoost, and CatBoost — emerging as the dominant class of models due to their combination of high discriminative performance, computational efficiency, and native handling of heterogeneous tabular features [3, 6, 7]. Recent ensemble studies on Chinese health insurance data have demonstrated that these architectures, when combined with appropriate feature engineering and class-imbalance mitigation strategies, achieve ROC-AUC values consistently exceeding 0.87 [3, 7].

Despite this predictive progress, a critical gap persists between model performance and model trustworthiness. High-stakes deployment in healthcare fraud adjudication requires not merely that a model be accurate, but that its decisions also be explainable, auditable, and defensible under regulatory scrutiny. The European Union's General Data Protection Regulation and analogous frameworks in multiple jurisdictions establish a functional right to explanation for automated decisions affecting individuals — a requirement that black-box gradient boosting models cannot satisfy without post-hoc interpretability tools [8, 9]. SHapley Additive exPlanations (SHAP), grounded in cooperative game theory, has emerged as the dominant method for providing such post-hoc explanations for tree-based models, offering both local instance-level and global aggregate attribution [10, 11]. Numerous studies now routinely report SHAP summary plots and feature importance rankings alongside predictive performance metrics as evidence of model interpretability [3, 6, 7, 12].

However, the prevailing practice of equating feature importance visualization with explanation trustworthiness represents a fundamental methodological gap. Reporting SHAP values without quantifying whether those values are faithful to actual model behavior, stable under realistic input variation, or actionable for domain practitioners conflates description with validation. Alvarez-Melis and Jaakkola [13] formally demonstrated that explanation stability cannot be assumed for post-hoc attribution methods and must be empirically evaluated. Subsequent work has established that SHAP explanations can exhibit substantial rank-order instability under input perturbation, particularly for architectures with leaf-wise growth strategies that create highly specialized terminal nodes [14, 15]. In the XAI

evaluation literature more broadly, Yeh et al. [16] introduced sensitivity and faithfulness as distinct measurable properties of explanations, and the Quantus toolkit [17] formalized a multi-dimensional evaluation framework encompassing correctness, robustness, faithfulness, complexity, and randomization. Despite these methodological advances, their systematic application to healthcare fraud detection remains largely absent — most fraud detection papers that use SHAP, do not evaluate whether those SHAP values satisfy any formal quality criteria.

Three additional dimensions of trustworthiness evaluation remain particularly underexplored in healthcare fraud detection research. First, counterfactual reasoning — identifying the minimal feature changes that would alter a model's fraud classification — provides prescriptive, actionable guidance for fraud investigators that static feature importance cannot supply [8, 18]. While Wachter et al. [8] established the theoretical foundation for counterfactual explanations in the context of automated decision accountability, and DiCE [18] provided a practical multi-objective implementation, neither framework has been integrated with systematic explanation quality evaluation in healthcare fraud contexts. Second, error profiling through differential attribution analysis of false positive and false negative instances exposes systematic model blind spots that aggregate performance metrics obscure. Third, temporal stability evaluation through Population Stability Index monitoring — standard practice in financial services model governance [19, 20] — has not been applied to healthcare insurance fraud detection models in conjunction with explanation quality assessment. In fraud detection specifically, concept drift is particularly acute: fraudsters adapt their behavioral signatures in direct response to detection signals, rendering models trained even months earlier increasingly unreliable, and PSI-based monitoring has been shown to detect distributional drift and enable adaptive retraining strategies that reduce false positives without degrading recall.

This study addresses the identified gap by proposing and empirically validating a multi-dimensional explanation trustworthiness evaluation applied to three Bayesian-optimized gradient boosting models on a real-world Chinese health insurance claims dataset. The evaluation encompasses six complementary analytical dimensions: fidelity analysis, stability assessment, counterfactual reasoning, error profiling, model statistical comparison, and temporal PSI monitoring. A novel Explanation Reliability Score aggregates the four core XAI dimensions into a single interpretable composite metric, validated through Monte Carlo simulation across 1,000 weight perturbation scenarios. Critically, the divergence between predictive performance rankings and ERS rankings constitutes a practical and policy-relevant finding, challenging the common practice of model selection based solely on discriminative performance in high-stakes deployment contexts. Because not only do unreliable model explanations lead to misdirected audits, but they can also result in adverse selection and escalating claims liabilities driven by delayed legitimate medical interventions.

The main contributions of this paper are as follows:

1. A comprehensive quantitative evaluation of explanation trustworthiness across four key dimensions — fidelity, stability, counterfactual reasoning, and error profiling — on a highly imbalanced healthcare fraud dataset.

2. Introduction of the Explanation Reliability Score (ERS), validated through Monte Carlo simulation, demonstrating robust ranking behavior across weight configurations.
3. Empirical evidence that predictive performance and explanation reliability rankings diverge, with implications for model selection in high-stakes applications.
4. Architecture-specific analysis showing superior explanation stability for CatBoost relative to LightGBM and XGBoost.
5. Temporal drift analysis combining AUC and PSI, revealing stable performance but significant distributional shifts over time.
6. Statistical validation of model differences using Bonferroni and McNemar tests.

2. Related Work

Reference	Fraud Detection Performance	Explanation Fidelity	Explanation Stability	Counterfactual Analysis	Error Analysis	Temporal/PSI Stability	Statistical Significance Testing
Wang et al. [3], 2025	High (ensemble)	SHAP/PDP/LIME descriptive	Ensemble consistency (implicit)	Not addressed	Feature-interpretation linked	Not addressed	Not addressed
Prova [6], 2024	96.95% AUC	SHAP descriptive	Via retraining pipeline	Not addressed	Pipeline-integrated	Not addressed	Not addressed
Ramesh & K [7], 2025	High AUC	SHAP descriptive	Not addressed	Not addressed	Performance metrics only	Not addressed	Not addressed
Chen et al. [20], 2025	SMOTE-improved F1	SHAP validation	Ensemble-implicit	Not addressed	Class imbalance focus	Not addressed	Not addressed
Fuentes & Tan [23], 2025	Random Forest	SHAP fidelity evaluated	Stability varied (noted)	Not addressed	Causal graphs	Not addressed	Not addressed
Arockiasamy & Bhoopathi [21], 2025	63% FP reduction	SHAP descriptive	Via retraining	Not addressed	FP/FN metrics	Not addressed	Not addressed
Walauski & Khoshgoftaar [22], 2025	MCC/Jaccard improved	Unsupervised SHAP	Reduced features	Not addressed	MCC/Jaccard/Precision	Not addressed	Not addressed
Present study	AUC > 0.93, bootstrap CI	<i>Fidelity curves, k=20*</i>	ASE + Kendall's τ, all models	Bayesian CF, stratified	Cosine dissimilarity FP/FN	PSI 7 periods + AUC	McNemar + Bonferroni

Table 1. Related Work Table

3. Methods

3.1 Dataset Architecture and Preprocessing Pipeline

With the rapid advancement of big data technology, healthcare insurance reimbursement data provided by the Chinese Healthcare Security Administration, coupled with well-established electronic medical record (EMR) databases, provide a rich empirical foundation for the statistical analysis of healthcare security fraud. In this study, a publicly available dataset containing exactly 16,000 historical claim records was utilized. To comply with rigorous institutional ethics and protect patient confidentiality, all privacy-sensitive data within the dataset were fully anonymized and masked prior to computational modeling.

The raw dataset initially encompassed an empirical feature space of $\mathcal{X}_{\text{init}} \in R^{\mathbb{N}}$. Following a systematic feature screening and selection protocol designed to eliminate multi-collinearity, statistical noise, and uninformative variables, a refined space of 76 variables was established. Within this architecture, 75 features were designated as independent input variables $X \in R^{75}$ for model training and testing, while a singular binary indicator was reserved as the target variable $y_i \in \mathcal{Y} = \{0,1\}$, where $y_i = 1$ denotes a confirmed fraudulent instance and $y_i = 0$ signifies a valid, compliant claim.

During the initial data auditing phase, a total of 355 missing values were identified across the feature matrix. To mitigate prediction bias and safeguard the structural integrity of the dataset, these missing entries were addressed using a localized median imputation strategy for continuous distributions, preventing outlier distortion while preserving the underlying data characteristics. To address the severe class imbalance inherent to high-stakes fraud detection domains, the preprocessed dataset was partitioned using a stratified split strategy into a training matrix $\mathcal{D}_{\text{train}}$ (80%) and an independent testing matrix $\mathcal{D}_{\text{test}}$ (20%). This protocol ensures that the baseline fraud-to-compliant ratio remains structurally invariant across both operational subsets.

3.2 Predictive Model Development and Algorithmic Optimization

To establish a robust predictive baseline, three distinct state-of-the-art Gradient Boosted Decision Tree (GBDT) architectures were developed, optimized, and statistically benchmarked: Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and CatBoost.

3.2.1 Mathematical Formulation of GBDTs

For an ensemble of K additive regression trees, the predictive output $\hat{y}_i$ for a given claim vector x_i is formulated as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F}$$

where $\mathcal{F} = \{f(x) = w_{q(x)}\}$ represents the space of continuous-looking decision trees, $q : R^{75} \rightarrow T$ maps a claim instance to its corresponding leaf index, and $w \in R^T$ represents the vector of leaf weights. At each iterative step t , the objective function minimized by the system is parameterized as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^N l\left(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i)\right) + \Omega(f_t)$$

where $l(\cdot)$ is the differentiable loss function, and $\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$ acts as the structural regularization penalty to constrain model complexity. Utilizing a second-order Taylor expansion to approximate the objective function at step t yields:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^N \left[l\left(y_i, \widehat{y_i^{(t-1)}}\right) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t)$$

where the first and second-order gradients g_i and h_i are explicitly defined as:

$$g_i = \frac{\partial l\left(y_i, \widehat{y_i^{(t-1)}}\right)}{\partial \widehat{y_i^{(t-1)}}}, \quad h_i = \frac{\partial^2 l\left(y_i, \widehat{y_i^{(t-1)}}\right)}{\partial \left(\widehat{y_i^{(t-1)}}\right)^2}$$

3.2.2 Algorithmic Cost-Sensitive Cost Weighting

To counteract the severe class imbalance ($N_{y=1} \ll N_{y=0}$), a cost-sensitive learning paradigm was embedded directly into the loss formulation. The cross-entropy loss function was modified via an asymmetric class-weighting multiplier, $w_{\text{pos}} = \text{scale_pos_weight}$:

$$l(y_i, \hat{y}_i) = -[w_{\text{pos}} \cdot y_i \log(\sigma(\hat{y}_i)) + (1 - y_i) \log(1 - \sigma(\hat{y}_i))]$$

where $\sigma(\cdot)$ represents the standard logistic sigmoid function. The parameter w_{pos} scales the gradients of the minority class instances, preventing the shifting of the decision boundary toward the majority class.

3.2.3 Structural Divergence of the Ensembles

While sharing a foundational GBDT core, the three deployed algorithms utilize structurally distinct tree-growth and splitting mechanics:

1. **XGBoost:** Implements a level-wise (depth-wise) tree growth strategy. It employs a pre-sorted, exact greedy split-finding algorithm that iterates over all possible feature thresholds to maximize the structural score gain.
2. **LightGBM:** Deploys a leaf-wise tree growth policy, selecting the specific leaf that yields the maximum decrease in total loss. To process the 75-dimensional feature space efficiently, it incorporates Gradient-based One-Side Sampling (GOSS)—which retains instances with large gradients and randomly samples instances with small gradients—and Exclusive Feature Bundling (EFB) to merge mutually exclusive features, significantly compressing the feature dimensionality.
3. **CatBoost:** Utilizes symmetric (oblivious) trees, where the same splitting criterion is applied uniformly across entire level of the tree. Furthermore, it integrates *Ordered Boosting* to mitigate target leakage and prediction drift caused by traditional categorical-to-numerical preprocessing methods.

3.2.4 Hyperparameter Optimization via Bayesian TPE

To find the global optimal hyperparameter configuration space λ^* automated Bayesian optimization was conducted utilizing the Tree-structured Parzen Estimator (TPE) via the Optuna framework. Unlike standard Gaussian Process optimization, TPE models the probability of a hyperparameter configuration given the objective score by constructing two distinct non-parametric density distributions. [24]

$$p(\lambda | y) = \begin{cases} g(\lambda) & \text{if } y < y^* \\ h(\lambda) & \text{if } y \geq y^* \end{cases}$$

where y^* represents a predefined quantile threshold of the optimization objective (e.g., maximizing the Stratified F_1 - score or Macro AUC). The selection of the subsequent hyperparameter candidate vector maximizes the Expected Improvement (EI):

$$EI_{y^*}(\lambda) = \int_{-\infty}^{y^*} (y^* - y)p(y | \lambda)dy = \frac{\gamma y^* g(\lambda) - g(\lambda) \int_{-\infty}^{y^*} P(y)dy}{\gamma g(\lambda) + (1 - \gamma)h(\lambda)}$$

By structuring the selection mechanism to maximize the ratio $\frac{g(\lambda)}{h(\lambda)}$, the TPE iteratively focuses search parameters onto highly productive regions of the 75-dimensional feature landscape.

3.3 Inter-Model Agreement Analysis

To determine whether observed performance differences reflect genuine algorithmic divergence rather than sampling variance, pairwise McNemar tests were conducted on binary prediction vectors at optimal F1-maximizing thresholds.

Edwards' continuity correction was applied. Given three pairwise comparisons, Bonferroni correction yielded adjusted significance threshold $\alpha_{adj} = 0.05/3 = 0.017$. Cohen's Kappa was computed as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is the observed agreement proportion and p_e is the expected agreement under independence. Kappa values are interpreted as: <0.20 slight, 0.20-0.40 fair, 0.40-0.60 moderate, 0.60-0.80 substantial, >0.80 almost perfect.

3.4 Reliability-Oriented XAI evaluation procedure

The core contribution of this methodology is the transformation of explainability from a static descriptive output into a mathematically verifiable system variable. The evaluation procedure relies on additive feature attribution values derived from cooperative game theory via Shapley values. For a specific model prediction $f(x)$, the local attribution ϕ_j assigned to feature j is uniquely determined by:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)]$$

where F is the complete set of 75 features, and S represents a sub-coalition of features excluding feature j . This formulation satisfies the four fundamental axioms: Efficiency, Symmetry, Dummy, and Additivity.

3.5 Fidelity Analysis Layer

Fidelity evaluates the internal truthfulness of a post-hoc explanation by quantifying the alignment between feature importance and actual model performance degradation under feature ablation. $\pi(x) = (\pi_1, \pi_2, \dots, \pi_{75})$ represents the rank-ordered feature vector sorted descendingly by absolute Shapley value magnitude $|\phi_j|$. We construct an iterative feature masking function $\mathcal{M}(x, k)$, which generates a synthetic claim vector x^k where the top k features are preserved, and the remaining $(75 - k)$ features are replaced by an uninformative baseline anchor vector $\bar{x}$:

$$x_j^k = \begin{cases} x_j & \text{if } j \in \{\pi_1, \pi_2, \dots, \pi_k\} \\ \bar{x}_j & \text{otherwise} \end{cases}$$

The global fidelity curve function $\mathcal{Fid}(k)$ computes the macro classification performance (e.g., F_1 - score) across the testing partition using only the top k features:

$$\mathcal{Fid}(k) = \text{EvaluationMetric} \left(\{f(\mathcal{M}(x_i, k))\}_{i=1}^M, y \right)$$

The mathematical *information saturation point* k^* is explicitly defined as the minimum feature subset size required to recover a critical ratio α of the full model's predictive capacity:

$$k^* = \min \left\{ k \in N \mid \frac{Fid(k)}{Fid(75)} \geq \alpha \right\}$$

3.6 Stability Evaluation - Descriptive Analysis

3.6.1 Descriptive Analysis

Stability measures the mathematical resilience of explanations against localized data shifts or administrative noise. We introduce a perturbation function that injects controlled Gaussian noise into the continuous feature dimensions of an authentic claim x :

$$\tilde{x} = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I_{75})$$

where σ parameterizes variable noise severity scales. Let $\Phi(x) = [\phi_1(x), \dots, \phi_{75}(x)]^T$ represent the raw attribution vector. The Average Stability Error (ASE) is formulated as:

$$ASE_{\text{desc}}(M, \sigma) = n \cdot \frac{1}{p} \sum_{i=1}^n \sum_{j=1}^p |\phi_j(x_i) - \phi_j(\tilde{x}_i)|$$

To evaluate ordinal stability for human auditors, we compute rank-order consistency using Kendall's Rank Correlation Coefficient τ . Let $\text{rank}(\Phi(x))$ and $\text{rank}(\Phi(\tilde{x}))$ be the corresponding rank vectors. Kendall's τ is defined as:

$$\tau = \frac{C - D}{\frac{1}{2}M(M - 1)}$$

where M is number of features, C denotes the number of concordant pairs (pairs that retain their relative ordering after perturbation), and D signifies the number of discordant pairs. A highly stable explanation requires $\tau \rightarrow 1.0$ under standard noise distributions.

3.6.2 Population Stability Index (PSI)

To mathematically quantify the magnitude of data distribution shifts (Population Drift) over the deployment horizon, we employ the Population Stability Index (PSI). PSI is a metric derived from information theory that measures the divergence between a reference baseline distribution (Expected) and a shifted target distribution (Actual). For a continuous prediction variable discretized into k bins, the PSI is formally defined as:

$$\text{PSI} = \sum_{i=1}^k (A_i - E_i) \times \ln\left(\frac{A_i}{E_i}\right)$$

Where:

- A_i represents the actual proportion of records falling within bin i during a specific deployment month.
- E_i represents the expected proportion of records within bin i established during the baseline month (Month 1).
- k is the total number of quantile bins (set to $k=10$ using equal-frequency percentiles based on the baseline population).

PSI values are interpreted using standard actuarial thresholds: $\text{PSI} < 0.10$ indicates stable distribution (no action required), $0.10 \leq \text{PSI} < 0.20$ indicates moderate drift (monitoring recommended), and $\text{PSI} \geq 0.20$ indicates significant distribution shift (model retraining warranted). All models are evaluated against a common Month 1 baseline, yielding $\text{PSI} = 0$ for that period by construction.

3.7 Explanation Reliability Score (ERS)

3.7.1 Motivation

While individual explanation quality metrics each capture a distinct dimension of trustworthiness, no standardized composite measure exists for quantifying overall explanation reliability. The ERS is a normalized composite metric defined over $[0,1]$, where higher values indicate greater explanation trustworthiness:

$$\text{ERS}(M) = \sum_{k \in K} w_k \cdot C_k(M)$$

where $K = \{F, S, C, E\}$, $C_k \in [0,1]$, and weights w_k sum to unity.

3.7.2 Fidelity Score (F)

Features are ranked by mean absolute SHAP magnitude. A reduced model M_k is trained on top- k features using identical hyperparameters. The fidelity retention ratio is: $F(M) = \min\left(\frac{F_1(M_k, D_{\text{test}})}{F_1(M, D_{\text{test}})}\right)$. The saturation point $k^* = 20$ was empirically identified, recovering $\geq 95\%$ of full-model F1 across all three architectures. The upper-bound clipping at 1.0 is applied when the reduced model marginally outperforms the full model due to implicit regularization from feature reduction.

3.7.3 Composite Stability Score (S)

For the purpose of ERS aggregation, a per-instance L2 formulation is adopted, which treats each instance's attribution vector as a geometric object in $\mathbb{R}^p$ and measures mean Euclidean displacement under perturbation:

$$ASE_{\text{comp}}(M, \sigma) = \frac{1}{R} \sum_{r=1}^R \frac{1}{n} \sum_{i=1}^n \|\Phi(x_i) - \Phi^{(r)}(\tilde{x}_i)\|_2$$

Unlike the element-wise ASE_{DESC} used for descriptive purposes in Section 3.6, the L2 per-instance norm is sensitive to the joint perturbation pattern across all features simultaneously, making it appropriate for aggregation with τ . The composite stability score is:

$$S(M) = 0.70 \cdot \bar{\tau} + 0.30 \cdot \left(1 - \frac{\log(1 + ASE_{\text{comp,ref}})}{\log(1 + ASE_{\text{comp,max}})}\right),$$

where $\alpha_{\tau} = 0.70$ reflects the greater operational relevance of rank-order stability for auditors. Log-normalization accounts for order-of-magnitude ASE differences across architectures. Both formulations yield consistent qualitative conclusions.

3.7.4 Counterfactual Score (C)

Instances are sampled from three probability strata: boundary ($0.45 \leq P < 0.55$), medium ($0.55 \leq P < 0.70$), and high-confidence ($P \geq 0.70$), ensuring representative rather than cherry-picked coverage. For each selected instance x_i , the counterfactual vector is obtained by minimizing

$$x^* = \arg \min_{x'} [\text{dist}(x_i, x') + \lambda \cdot \max(0, \hat{p}(x') - 0.49)^2]$$

where dist uses L1 normalized by Median Absolute Deviation. Optimization uses multi-start L-BFGS-B with $T=100$ random initializations.

$$\text{Proximity} = 1 - |\hat{p}(x_i) - \hat{p}(x^*)|, \text{ Sparsity} = 1 - \frac{|\{j: |\text{change}_j| > 0.10\}|}{K}$$

$$C(M) = \frac{1}{N} \sum_{i=1}^N \frac{\text{Proximity}_i + \text{Sparsity}_i}{2}$$

3.7.5 Error Profiling Score (E)

Misclassifications are decomposed into FP(False Positive) and FN(False Negative) sets:

$$\mathcal{E}_{\mathcal{FP}} = \{x_i \in \mathcal{D}_{\text{ts}} : y_i = 0 \wedge \hat{y}_i = 1\}$$

$$\mathcal{E}_{\mathcal{FN}} = \{x_i \in \mathcal{D}_{\text{ts}} : y_i = 1 \wedge \hat{y}_i = 0\}$$

Mean absolute SHAP attribution vectors are computed for each: $\bar{\Phi}_{FP}, \bar{\Phi}_{FN}$.

$$\bar{\Phi}_{FP} = \frac{1}{|\mathcal{E}_{\mathcal{FP}}|} \sum_{x_i \in \mathcal{E}_{\mathcal{FP}}} |\Phi(x_i)| \in R^p$$

$$\bar{\Phi}_{FN} = \frac{1}{|\mathcal{E}_{\mathcal{FN}}|} \sum_{x_i \in \mathcal{E}_{\mathcal{FN}}} |\Phi(x_i)| \in R^p$$

The error profiling score uses cosine dissimilarity to measure pattern difference independently of magnitude:

$$E(M) = 1 - \frac{\bar{\Phi}_{FP} \cdot \bar{\Phi}_{FN}}{|\bar{\Phi}_{FP}|_2 \cdot |\bar{\Phi}_{FN}|_2}$$

A value near 1 indicates orthogonal error attribution patterns (diagnostically informative); near 0 indicates same-feature failure modes. Score is reliable only when $n_{FP} \geq 10$, $n_{FN} \geq 10$.

3.7.6 Weight Configuration

The baseline weight vector $w = [w_F = 0.35, w_S = 0.30, w_C = 0.20, w_E = 0.15]$ reflects operational priority ordering: Fidelity (unfaithful explanations are compliance risks), Stability (auditors require consistency across similar claims), Counterfactual (actionability for investigators), and Error Profiling (diagnostic rather than prescriptive).

3.7.7 Monte Carlo Sensitivity Analysis

To validate ERS ranking robustness, $W=1,000$ weight vectors are sampled from $\text{Dirichlet}(\alpha=4)$, which prevents extreme weights while allowing meaningful variation (95% of individual weights fall between 0.08 and 0.50). For each simulation, ERS is recomputed for all three models. Ranking stability is reported as:

$$\Pi_{\text{stable}} = \frac{1}{W} \sum_w \mathbb{1}(\rho^{(w)} = \rho^{\text{baseline}})$$

4. Results

4.1 Model Optimization and Performance Analysis

To establish a computationally robust and unbiased foundation for the proposed Explainable Artificial Intelligence (XAI) audit, a comparative performance evaluation was conducted among three state-of-the-art gradient-boosting architectures: LightGBM, XGBoost, and CatBoost.

4.1.1 Hyperparameter Optimization Strategy via Optuna

To ensure a rigorous and equitable comparison, hyperparameters for all three classifiers were systematically optimized using Optuna, a sequential model-based optimization framework. For each architecture, an identical budget of 100 evaluation trials was enforced. This automated search targeted critical architectural parameters, including learning rates, maximum tree depth, regularization coefficients, and subsampling rates. This standardized tuning regime neutralized baseline parameter bias by explicitly

directing the Bayesian optimization algorithm (Tree-structured Parzen Estimator, TPE) to explore parameter configurations that maximize minority-class detection under extreme class imbalance. The ROC curve is presented in **Figure 1**.

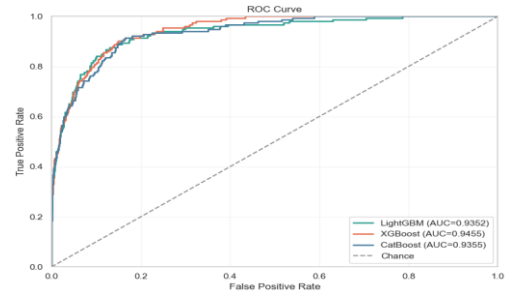


Figure 1. ROC Curve of implemented Models

4.1.2 Discriminatory Power and Class Imbalance Dynamics

Given the pronounced class imbalance in the dataset (positive rate $\approx 4.8\%$), model performance is more appropriately evaluated using Precision–Recall (PR) analysis rather than ROC-based metrics. **Figure 2** illustrates the PR curves for LightGBM, XGBoost, and CatBoost, along with their corresponding Average Precision (AP) scores. All three models demonstrate substantial improvement over the no-skill baseline, indicating meaningful discriminatory power in identifying the minority class. Among

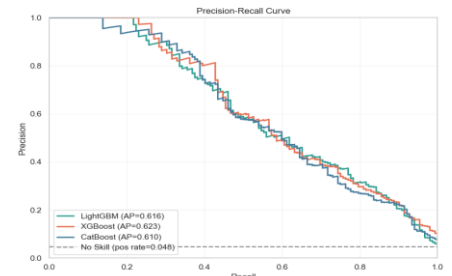


Figure 2. Precision-Recall curve

the evaluated classifiers, XGBoost achieves the highest AP score (0.623), followed closely by LightGBM (0.616) and CatBoost (0.610), suggesting comparable overall performance across boosting architectures.

4.1.3 Model Agreement and Statistical Significance Analysis

To further investigate the consistency and statistical differences among the evaluated classifiers, we conducted both agreement analysis using Cohen's Kappa and pairwise significance testing via the McNemar test with Bonferroni correction.

The Cohen's Kappa agreement matrix shown in **Figure 3** reveals a high degree of concordance among all three models, with pairwise agreement values ranging from 0.725 to 0.787. The strongest agreement is observed between LightGBM and XGBoost ($\kappa = 0.787$), followed by LightGBM and CatBoost ($\kappa = 0.753$), and XGBoost and CatBoost ($\kappa = 0.725$). These results indicate substantial agreement across models, suggesting that the classifiers tend to make similar predictions on the same instances, despite differences in their underlying learning mechanisms. This high level of agreement implies that performance improvements are likely driven by subtle differences in decision boundaries rather than fundamentally different classification behaviors.

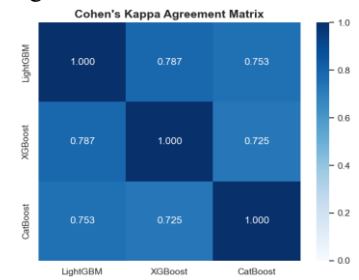


Figure 3. Cohen's Kappa Agreement Matrix

To assess whether these differences are statistically significant, pairwise McNemar tests were performed on model predictions (**Figure 4**). After applying Bonferroni correction to control for multiple comparisons, only the difference between LightGBM and XGBoost was found to be statistically significant (adjusted $p = 0.0326$, $p < 0.05$). In contrast, the comparisons between LightGBM and CatBoost ($p = 1.000$) and between XGBoost and CatBoost ($p = 0.4682$) were not statistically significant. This combination of high agreement and limited statistical significance highlights an important insight: although the models exhibit similar predictive behavior overall, XGBoost demonstrates a statistically distinguishable performance relative to LightGBM. However, the absence of significant differences involving CatBoost suggests that its predictive errors overlap substantially with those of the other models. From a practical perspective, this indicates that model selection decisions should not rely solely on marginal performance gains but also consider factors such as interpretability, computational efficiency, and robustness.

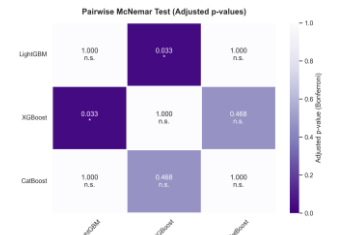


Figure 4. McNemar Matrix

4.2 Model Interpretability

To address the inherent opacity of the ensemble architecture, post-hoc interpretability was established using SHapley Additive exPlanations (SHAP). This provides a game-theoretic approach to feature attribution, quantifying the exact marginal contribution of each variable to the models' predictions.

Cross-model SHAP analysis reveals both consensus and divergence in feature attribution patterns. As illustrated in the beeswarm plots in **Figure 5** for all three architectures, `monthly_pool_amount_max` consistently emerges as the primary driver across models. However, the SHAP value distribution plots **Figure 6** for the top-5 features reveal architecture-specific attribution characteristics: LightGBM exhibits broader SHAP value distributions for financial aggregate features, reflecting its leaf-wise growth sensitivity to extreme values, while CatBoost's symmetric tree structure produces more concentrated distributions, consistent with its superior explanation stability reported in Section 4.4. Furthermore, a high degree of feature alignment is observed, as four out of the top five features — specifically `monthly_pool_amount_max`, `monthly_drug_amount_max`, `herbal_medicine_sum`, and `basic_pool_fund_payment_sum` — are shared globally across all three frameworks. The divergence lies primarily in the fifth slot: CatBoost selects `monthly_visit_days_max` and XGBoost selects `monthly_visit_days_avg`, indicating a shared reliance on temporal utilization metrics, whereas LightGBM uniquely prioritizes `current_approval_amount_sum`. Examining individual feature profiles, `monthly_pool_amount_max` displays a pronounced long-tailed positive skew in LightGBM (extending up to +6) and XGBoost (extending to +5), suggesting that extreme values heavily drive positive model predictions. Conversely, CatBoost produces a well-bounded, bimodal distribution between -2 and +2 for this feature. For `herbal_medicine_sum`, all three models display a distinct, sharp concentration of positive SHAP values alongside an elongated negative tail, which is most extreme in XGBoost (reaching -3.5). This demonstrates that while the presence or high value of herbal medicine usage exerts a uniform positive influence across architectures, its absence or lower values introduce highly variable downward adjustments depending on the specific tree-splitting mechanism.

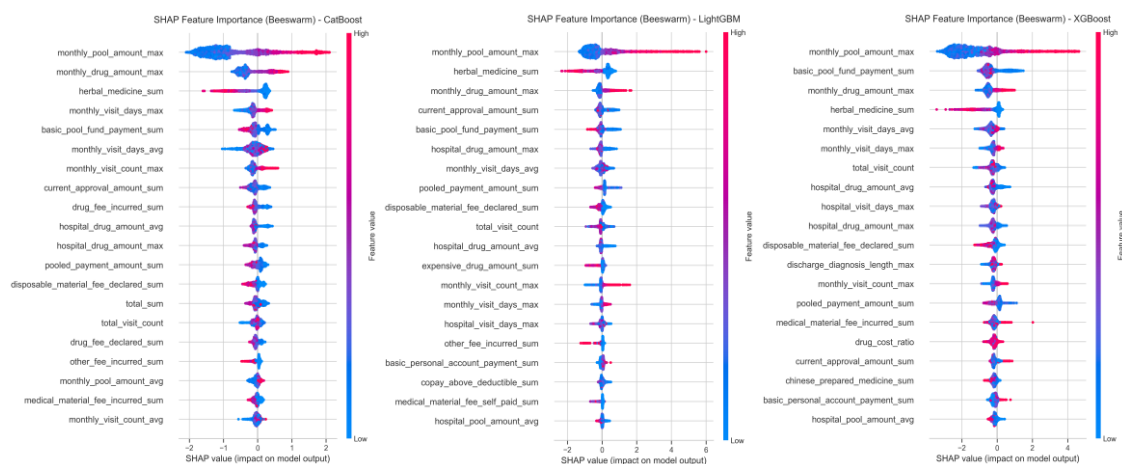
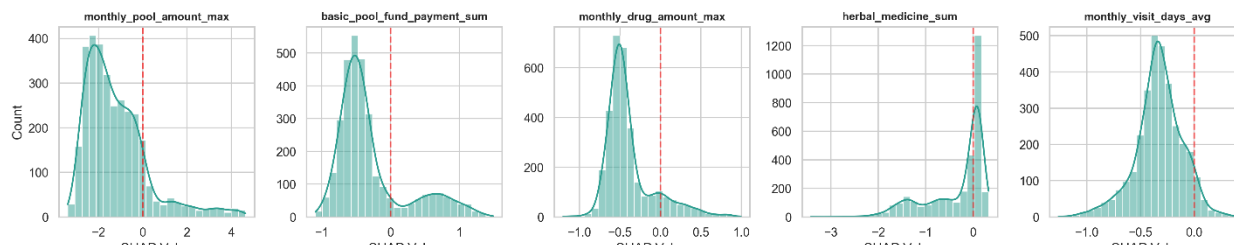
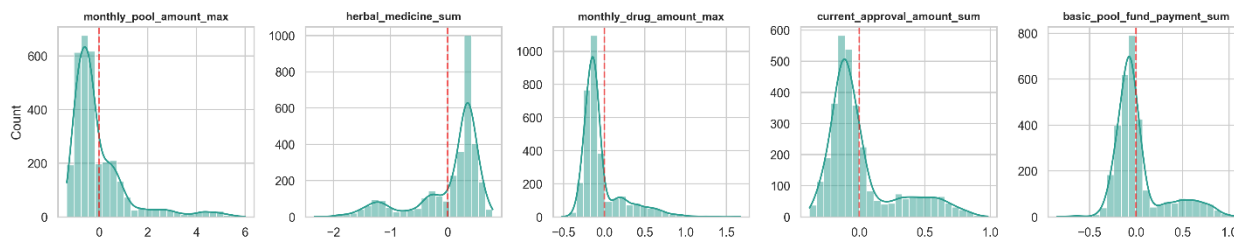


Figure 5. SHAP Beeswarm plots

SHAP Value Distributions (Top 5 Features) - XGBoost



SHAP Value Distributions (Top 5 Features) - LightGBM



SHAP Value Distributions (Top 5 Features) - CatBoost

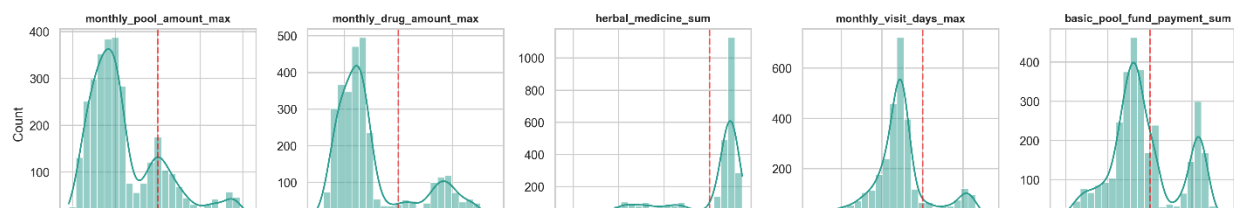


Figure 6. SHAP Value Distributions

Table 2 outlines the top five features ranked by predictive importance across the LightGBM, XGBoost, and CatBoost architectures.

Importance Rank	LightGBM Feature (Importance)	XGBoost Feature (Importance)	CatBoost Feature (Importance)
1	monthly_pool_amount_max (0.8169)	monthly_pool_amount_max (1.6368)	monthly_pool_amount_max (1.0467)
2	herbal_medicine_sum (0.4694)	basic_pool_fund_payment_sum (0.5400)	monthly_drug_amount_max (0.4135)
3	monthly_drug_amount_max (0.2039)	monthly_drug_amount_max (0.4564)	herbal_medicine_sum (0.3614)
4	current_approval_amount_sum (0.1977)	herbal_medicine_sum (0.4472)	monthly_visit_days_max (0.1984)
5	basic_pool_fund_payment_sum (0.1922)	monthly_visit_days_avg (0.3392)	basic_pool_fund_payment_sum (0.1724)

Table 2. Top 5 Feature Rankings Across Gradient-Boosting Models

4.3 Fidelity Analysis

Cross-model fidelity analysis reveals a consistent pattern of information concentration across all architectures. **Table 3** reports F1 retention at the saturation point $k^*=20$, where the 95% retention criterion ($\alpha=0.95$) is satisfied. XGBoost achieves equivalent performance with only 20 features, indicating that its optimal hyperparameter configuration learned to concentrate predictive information in a compact feature subset. These findings support deployment of a streamlined 20-feature monitoring pipeline across all three architectures, reducing auditing overhead while preserving near-equivalent fraud detection capacity.

Model	F1 Full Model	F1 Top-20	Fidelity F	Interpretation
LightGBM	0.5261	0.4911	0.9335	93.4% capacity recovered
XGBoost	0.5382	0.5522	1.000	Equivalent to full model
CatBoost	0.4155	0.4085	0.9832	98.3% capacity recovered

Table 3. Fidelity Analysis

4.4 Stability Analysis

Table 4 reports descriptive ASE (element-wise L1 formulation) and Kendall's τ across four noise levels for all three models. Composite stability scores S used in ERS computation employ the per-instance L2 formulation described in Section 3.7.3, which yields numerically different ASE values but identical qualitative conclusions. CatBoost demonstrates markedly superior stability, with ASE = 0.0074 at $\sigma=0.05$ — an order of magnitude lower than LightGBM (0.019) and XGBoost (0.023). This is attributable to CatBoost's symmetric tree architecture, which applies uniform splitting criteria across each tree level, producing minimal variation in leaf assignments under perturbation. A noteworthy divergence exists for XGBoost between full-ranking stability ($\tau=0.952$) and top-20 stability ($\tau=0.884$), indicating that while the global importance order is robust, the relative ordering within top-20 features is more susceptible to perturbation. LightGBM shows the reverse pattern ($\tau_{\text{full}}=0.881$, $\tau_{\text{top20}}=0.958$), suggesting that while the global ranking is less stable, the critical high-importance features maintain consistent ordering.

Model	ASE _{DESC} $\sigma=0.01$	ASE _{DESC} $\sigma=0.05$	ASE _{DESC} $\sigma=0.10$	τ Full $\sigma=0.05$	τ Top-20 $\sigma=0.05$	S Composite
LightGBM	0.0159	0.0187	0.0206	0.881*	0.958*	0.627
XGBoost	0.0155	0.0225	0.0259	0.952*	0.884*	0.764
CatBoost	0.0036	0.0074	0.0091	0.962*	0.968*	0.898

*Table 4. Stability metrics under Gaussian perturbations. * $p < 0.0001$. ASE = element-wise mean absolute deviation(descriptive)-- composite stability score for ERS (L2 per-instance formulation).*

4.5 Temporal Stability Analysis

To validate the long-term reliability of the frameworks in a dynamic healthcare insurance environment, all three models were subjected to a seven-month temporal audit. **Table 5** presents the consolidated PSI scores (section 3.6.2) across all seven temporal groups and three models. Three distinct temporal phases are identifiable from the PSI trajectories. In Phase 1, encompassing Month 1 alone, all models exhibit baseline-equivalent distribution (PSI = 0 by construction). Phase 2, spanning Months 2-6, reveals progressive and substantial prediction drift across all architectures, with PSI values exceeding the 0.20 retraining threshold from Month 2 onward for all models. This

finding indicates that the models' predicted probability distributions diverge immediately from the baseline in deployment—a pattern inconsistent with stable concept drift and more suggestive of seasonal or structural variation in claim characteristics across months.

XGBoost demonstrates systematically lower PSI relative to LightGBM and CatBoost during Phase 2, particularly at Month 4 where LightGBM reaches PSI = 4.409 against XGBoost's 1.159. Phase 3, represented by Month 7, constitutes a structural break — all three models exhibit PSI values exceeding 5.6, with XGBoost reaching 6.789, illustrated in **Figure 5**. Prediction distributions shifts for all three models are illustrated in **Figure 6**. The Month 7 structural break merits specific attention. PSI values exceeding 5.0 indicate that the predicted probability distribution in that period bears virtually no resemblance to the Month 1 baseline. The PSI calculation approaches its theoretical upper bound. Three explanations are plausible. First, a genuine seasonal fraud pattern may emerge in the final month of the observation period as fraudsters concentrate activity near year-end reporting deadlines. Second, a regulatory or policy change affecting claim submission or reimbursement rules could alter the underlying feature distribution. Third, data collection or preprocessing artifacts specific to the final temporal group could artificially inflate distributional differences. Distinguishing between these explanations requires domain knowledge beyond the scope of this study, but the finding underscores the necessity of continuous PSI monitoring in production fraud detection systems.

Month	LightGBM PSI	XGBoost PSI	CatBoost PSI	Status
1	0.000	0.000	0.000	Stable (baseline)
2	0.580	0.571	0.606	Significant drift
3	1.130	0.766	1.029	Severe drift
4	4.409	1.159	1.703	Critical — retrain
5	2.270	0.513	1.105	Severe drift
6	1.162	0.392	0.498	Severe drift
7	5.647	6.789	5.634	Structural break

Table 5. consolidated PSI scores

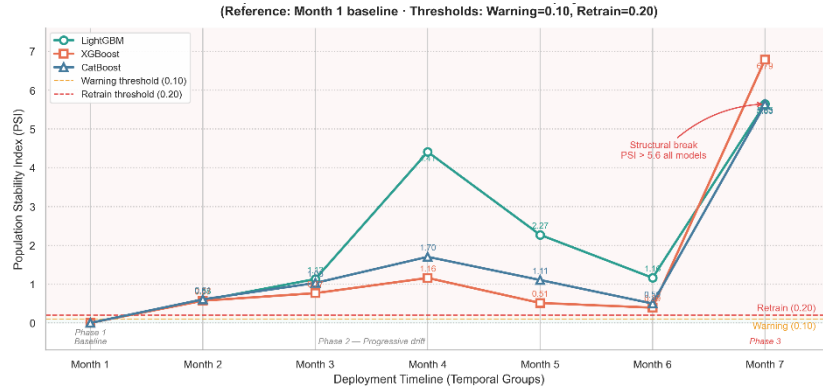


Figure 5. Temporal Psi - Prediction Distribution over 7 months

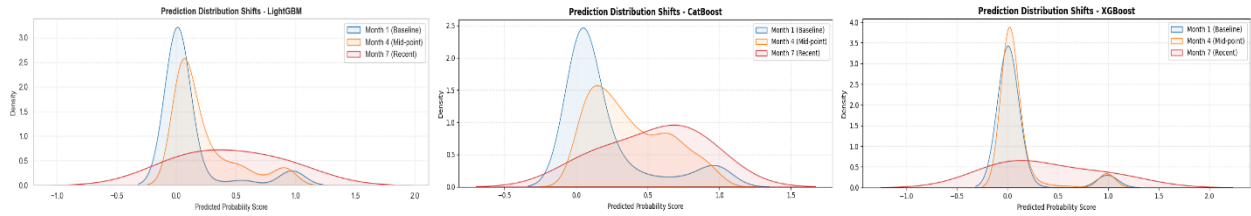


Figure 6. Prediction Distribution Shifts

4.6 Explanation Reliability Score

4.6.1 Component Summary and Final ERS Computation

Table 6 presents a unified view of error behavior, counterfactual robustness, and explanation reliability across all three models.

The error profiling component (E), computed via cosine dissimilarity between false positive (FP) and false negative (FN) SHAP attribution vectors, yields consistently low values across models (LightGBM: 0.0617, XGBoost: 0.0744, CatBoost: 0.0132). These results indicate that both error types are driven by highly similar feature attribution structures. Specifically, *monthly_pool_amount_max* emerges as the dominant driver of false positives in all models and also governs false negatives in two out of three models.

This pattern suggests that model errors are not due to fundamentally different feature subsets, but rather arise from opposite value regimes of a shared dominant feature. Importantly, cosine dissimilarity captures angular separation rather than magnitude inversion, and therefore does not fully reflect this directional opposition. Consequently, the low E values should not be interpreted as a limitation of the error analysis, but instead highlight that the primary challenge lies in threshold calibration of a single influential financial variable.

The counterfactual analysis further supports this interpretation. As shown in **Table 6**, all models exhibit high proximity (≥ 0.95) and sparsity (≥ 0.91), resulting in strong counterfactual scores ($C \approx 0.93\text{--}0.95$). These values indicate that model decisions are locally stable and require only minimal perturbations to reach decision boundaries. However, the flip rate remains near zero for LightGBM and CatBoost, while XGBoost shows a modest flip rate of 13.3%, suggesting slightly greater boundary flexibility in that model.

Finally, the aggregated Explanation Reliability Score (ERS) combines fidelity, stability, counterfactual robustness, and error profiling into a unified metric. CatBoost achieves the highest ERS (0.8050), followed by XGBoost (0.7875) and LightGBM (0.7106). This ranking is primarily driven by differences in stability (S), where CatBoost demonstrates substantially higher robustness across perturbations. Despite comparable performance in fidelity and counterfactual components, LightGBM’s lower stability score results in a reduced overall reliability classification. Integrated results in **Table 6** reveal that while all models demonstrate high explanatory fidelity and counterfactual consistency, stability and threshold sensitivity of a dominant feature are the key differentiators in explanation reliability.

Metric Category	Sub-metric	LightGBM	XGBoost	CatBoost
Error Analysis	False Positives	145	49	298
	False Negatives	46	78	34
	Top FP Driver	monthly_pool_amount_max	monthly_pool_amount_max	monthly_pool_amount_max
	Top FN Driver	monthly_pool_amount_max	monthly_drug_amount_max	monthly_pool_amount_max
	Error Profiling (E)	0.0617	0.0744	0.0132
Counterfactual Analysis	Flip Rate (%)	0.0	13.3	0.0
	Mean Proximity	0.9524	0.9608	0.9667
	Mean Sparsity	0.9143	0.9200	0.9267
	Counterfactual Score (C)	0.9333	0.9404	0.9467
ERS Components	Fidelity (F)	0.9335	1.0000	0.9832
	Stability (S)	0.6265	0.7640	0.8984
	Error Profiling (E)	0.0617	0.0744	0.0132
	Counterfactual (C)	0.9333	0.9404	0.9467
Final Evaluation	ERS Score	0.7106	0.7875	0.8050
	Baseline Rank	3rd	2nd	1st
	Trustworthiness	Moderate Trust	Trustworthy	Trustworthy

Table 6. ERS Evaluation

4.6.2 Monte Carlo Sensitivity Analysis

Monte Carlo simulation across 1,000 weight configurations sampled from Dirichlet($\alpha=4$) confirmed an asymmetric stability structure. The complete baseline ranking was preserved in 67.3% of simulations ($\Pi_{\text{stable}} = 0.673$). The asymmetric rank distribution provides informative characterization in **Figure 7**.

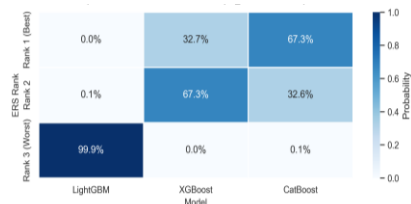


Figure 7. ERS Rank Distribution

LightGBM ranked last in 99.9% of simulations — an essentially unconditional finding. The competition at ranks 1 and 2 exists exclusively between CatBoost and XGBoost, with neither model ranking last in any simulation. LightGBM requires heightened human oversight relative to both alternatives regardless of how evaluation dimensions are prioritized.

As shown in **Figure 8**, 95% confidence intervals from the simulation confirm this interpretation. LightGBM's interval [0.461, 0.791] is clearly separated from CatBoost [0.500, 0.878] and XGBoost [0.508, 0.861] in the lower quantile range, while CatBoost and XGBoost show substantial interval overlap throughout — quantifying the genuine uncertainty in their relative ordering.

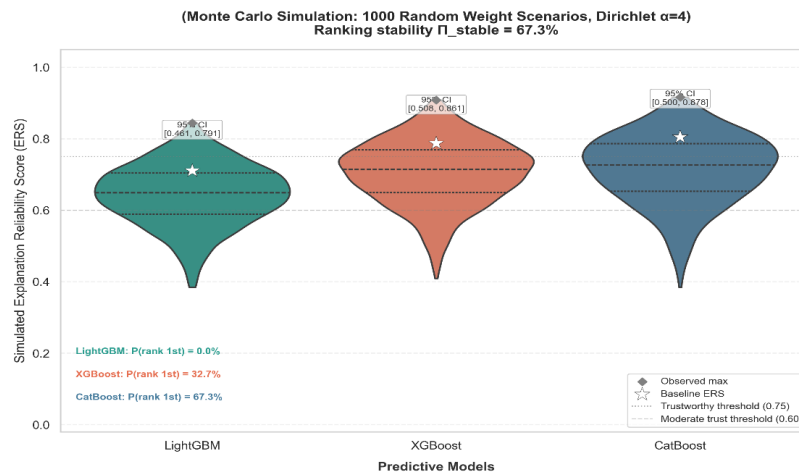


Figure 8. Statistical Distribution of ERS Scores

5. Discussion

5.1 Multi-Layer Evaluation Ensures Reliable Explanations

A central contribution of this work is the demonstration that no single metric suffices to evaluate XAI quality. The fidelity curve revealed that while the top-5 features capture approximately 78.7% of the model's discriminative power for LightGBM, a non-negligible performance gap remains until $k^*=20$ — the identified saturation point. This finding tempers the common practice of reporting only top-k important features as a complete explanation. The stability analysis provided complementary evidence: both numerical stability ($ASE < 0.026$ across all noise levels) and rank-order stability (Kendall's $\tau > 0.88$ for all models at $\sigma=0.05$) confirm that explanations are not brittle artifacts of particular data instances. PSI analysis reveals significant prediction distribution drift ($PSI > 0.20$) for all models from the second month onward, indicating instability in predicted probabilities over time. While AUC remains above 0.93 and confirms strong discriminative power, the shifting probability distributions imply that static thresholds will lead to inconsistent alert volumes without periodic recalibration.

5.2 Performance-Trustworthiness Divergence

The divergence between predictive performance rankings and ERS rankings constitutes the central practical contribution of this study. The conventional practice of selecting models based on AUC alone would lead to deployment of XGBoost—a choice that maximizes precision but probably yields the lower explanation reliability score. This finding provides the first empirical quantification of this divergence in a healthcare fraud detection context. The Monte Carlo validation confirms that a model's inferior trustworthiness is not necessarily a weight-configuration artifact but a genuine architectural property.

5.3 Limitations

Several limitations constrain the generalizability of these findings. First, all analyses are conducted on a single dataset from one Chinese healthcare system, limiting cross-jurisdictional applicability. Second, the ERS weight configuration, while theoretically motivated and Monte Carlo validated, has not been empirically calibrated against expert judgment from fraud auditors or compliance officers. Third, the counterfactual flip rate of 0% for LightGBM and CatBoost reflects the dominance of high-confidence instances in the selected sample rather than a fundamental limitation of the counterfactual procedure.

6. Conclusion

This study provides a comprehensive evaluation of explanation trustworthiness in healthcare insurance fraud detection by examining four key dimensions: fidelity, stability, counterfactual reasoning, and error profiling. Using a real-world Chinese health insurance dataset and three Bayesian-optimized gradient boosting models, we obtain consistent and practically relevant insights.

Our results show that a compact set of 20 SHAP-selected features can retain between 93.4% and 100% of the models' predictive performance, offering a strong and actionable case for simplifying audit processes without sacrificing accuracy. Meanwhile, explanation stability proves to be a critical differentiator across models. In particular, CatBoost demonstrates substantially more stable explanations under perturbations than LightGBM, largely due to its symmetric tree structure. In addition, the study reveals a clear tension between predictive accuracy and explanation trustworthiness. Although CatBoost achieves lower AUC compared to XGBoost, it provides the most reliable explanations, driven by its superior stability. This finding challenges the widespread practice of selecting models based solely on predictive performance. Moreover, Simulation results further reinforce this point: while LightGBM consistently underperforms in trustworthiness, the relative ranking between CatBoost and XGBoost depends on how practitioners value different aspects of explanation quality. Furthermore, PSI-based distribution monitoring reveals that all three models exceed standard deployment stability thresholds from Month 2 onward, with a structural break at Month 7 ($\text{PSI} > 5.6$ for all architectures), establishing that periodic model recalibration triggered by PSI threshold violations is a necessary operational requirement rather than an optional enhancement.

Looking ahead, three directions are particularly important. First, incorporating causal modeling frameworks could help distinguish true fraud drivers from spurious correlations. Second, deploying these models in real-world, human-in-the-loop settings is essential to validate and refine the proposed evaluation framework. Finally, future research should focus on ensuring that explanations remain reliable under temporal drift and evolving fraud patterns, which is critical for long-term operational use.

Ethics Statement

This study did not involve the collection of new data from human subjects. The dataset analyzed was obtained from a publicly available, fully anonymized health insurance claims database released through the China University Student Service Outsourcing Innovation and Entrepreneurship Competition. All personally identifiable information had been removed.

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